Elements of AIML Lab
Experiment 02 (After-submission Reference): WEKA for Classification

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1 How to use this reference

This document is meant to be shared **after you submit** Experiment 02. Use it to verify that you:

- selected a **nominal** class attribute (label),
- evaluated models fairly (same split/CV),
- interpreted the confusion matrix and precision/recall/F1 correctly.

2 Quick recap: what classification means

Classification is **supervised learning** where the target is a **label** (nominal), e.g., spam/ham, pass/fail, disease/no-disease.

2.1 Confusion matrix (binary)

	Pred +	Pred -
True +	TP	FN
True -	FP	TN

Interpretation in plain words:

- TP: correct "positive" predictions
- FP: false alarms
- FN: missed positives
- TN: correct "negative" predictions

2.2 Metrics (why more than accuracy)

- Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$ (can be misleading if classes are imbalanced).
- Precision $= \frac{TP}{TP+FP}$ (quality of positive predictions).
- Recall $= \frac{TP}{TP+FN}$ (coverage of true positives).
- F1 balances precision and recall.

3 Worked example in WEKA (end-to-end)

3.1 Dataset choice (example)

WEKA often provides sample classification datasets under a `data/` folder (e.g., `iris.arff`, `weather.nominal.arff`). Your screenshots/metrics may differ depending on dataset choice.

3.2 Step 1: load data and set class attribute

1. WEKA → **Explorer** → **Preprocess** → **Open file...**
2. Set **Class** to the label column.

Cross-check: the class attribute must be **nominal**. If numeric, you are doing regression.

3.3 Step 2: (optional) Replace missing values and remove IDs

- Missing values: `Filter` → `unsupervised/attribute/ReplaceMissingValues`.
- Remove ID-like columns (roll number, customer ID) that leak identity rather than signal.

3.4 Step 3: train and evaluate two classifiers

Model A: Decision tree (J48).

1. **Classify** tab → **Choose** → `trees/J48`.
2. **Test options** → **Cross-validation** (10 folds) *or* **Percentage split**.
3. Click **Start**.

Model B: Naive Bayes (a fast baseline).

1. **Choose** → `bayes/NaiveBayes`.
2. Keep the **same** test option as Model A.
3. Click **Start**.

3.5 Step 4: interpret WEKA output

In the output, you typically see:

- overall accuracy,
- confusion matrix,
- per-class precision/recall/F1,
- weighted averages (important when classes are imbalanced).

Practical reading tip: If FN is costly (e.g., missing a disease), focus on **recall**. If FP is costly (e.g., blocking a genuine payment), focus on **precision**.

4 What a strong submission looks like (cross-check)

4.1 Screenshots

- Preprocess: correct class selected.
- Classify: model name + evaluation method.
- Output: confusion matrix + precision/recall/F1 for both models.

4.2 Report structure (1–2 pages)

1. Dataset and class labels (what does "positive" mean?).
2. Preprocessing steps (if any).
3. Two models, same evaluation method.
4. Results table (accuracy, precision, recall, F1).
5. Observations (trade-offs: which model increases recall but decreases precision, etc.).

5 Troubleshooting (common issues and fixes)

- **Wrong class attribute:** selecting an ID column creates meaningless results.
- **Different evaluation settings across models:** comparisons become unfair.
- **Only reporting accuracy:** always include confusion matrix + precision/recall/F1.
- **Imbalance ignored:** if one class is rare, weighted averages and minority-class recall matter most.

Sources: Provost and Fawcett, *Data Science for Business*.;McKinney, *Python for Data Analysis*.;WEKA Documentation (Waikato Environment for Knowledge Analysis), accessed 2026-02-28.;Matplotlib and Seaborn documentation, accessed 2026-02-28.